

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

2nd Semester of B. Pharm. Examination

University Theory Examination September 2018

PH119/127 Pharmaceutical Chemistry-II

Date: 21.09.2018, Friday

Time: 10:00 a.m. to 01:00 p.m.

Maximum Marks: 80

Instructions:

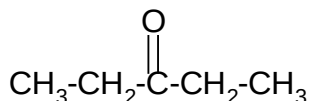
1. There are three sections in this question paper.
 2. SECTION – I comprises of Question 1. Total marks for Section 1 are 20. There are 20 sub-questions (MCQ type). Answers to SECTION – I are to be given in Answer Sheet for MCQ type questions provided to you. Maximum time allotted for SECTION – I is 30 minutes. Answers to SECTION – I must be written during the first 30 minutes of the examination.
 3. Answers to SECTION – II and SECTION – III are to be provided in separate Main Answer Books provided to you.
 4. Figures to right indicate marks.
 5. Draw neat sketches wherever necessary.
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SECTION - I

Q 1 Attempt all questions. Each question is of one mark.

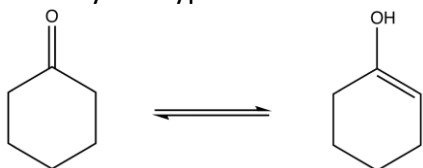
20

1. All are correct about Carbanion; Except One
[A] EWG increase the stability of carbanion
[B] The stability order is Primary > Secondary > Tertiary
[C] Carbanion stability can be increase if it is attached with aromatic system
[D] EDG increase the stability of carbanion
2. The IUPAC name of following compound is



- [A] 3-Pentanone
[B] 3-Butanone
[C] 3-Keto Butane
[D] 3-Oxo Pentanone
3. The IUPAC name of neopentane is
[A] 1,2 dimethyl propane
[B] 2,2 dimethyl propane
[C] 3-methyl butane
[D] 2-methyl butane
4. All statement is correct for SN₂ reaction. Except one
[A] Follow Second order kinetic
[B] Rearrangement is possible
[C] Inversion of configuration takes place
[D] Single step reaction

5. In the molecule of Alkene, what is the hybridization?
 [A] sp^3
 [B] sp^2
 [C] Unhybrid
 [D] sp
6. Unit of Dipole Moment is: _____
 [A] Cm^{-1}
 [B] Poise
 [C] Debye
 [D] ppm
7. Which of the following has the highest electronegativity
 [A] Br
 [B] Cl
 [C] F
 [D] I
8. Orientation of Elimination reaction follows...
 [A] Markoniov's rule
 [B] Saytzeff rule
 [C] Micheal addition
 [D] Hoffmann rule
9. Alkyl halides undergo a type of reaction
P. Nucleophilic substitution, **Q.** Nucleophilic addition
R. Elimination, **S.** Electrophilic aromatic substitution
 [A] P and Q are correct
 [B] P and R are correct
 [C] P,Q, R and S are correct
 [D] Q and R are correct
10. Identify the types of Tautomerism observed in following figure.



- [A] Amine Imine
 [B] Keto Enol
 [C] Lactam Lactim
 [D] Amide Imidic
11. Identify the following type's dienes.



- [A] Conjugated Diene
 [B] Isolated Diene
 [C] Cumulated Diene
 [D] None of above
12. The halogenation of alkane is based on formation of
 [A] Carbocation
 [B] Carbanion
 [C] Free radical
 [D] None

13. During elimination reaction of halogenoalkanes reagent use is
[A] Ethanolic sodium hydroxide
[B] Ethanoic acid
[C] Ethanoate
[D] Ethane
14. When the nucleophile :OR attacks the RX, the resultant product will be
[A] ROH
[B] ROR
[C] RCN
[D] RNHR
15. Which of the following cannot react as a nucleophile?
[A] CH_3NH_2
[B] $(\text{CH}_3)_2\text{NH}$
[C] $(\text{CH}_3)_3\text{N}$
[D] $(\text{CH}_3)_4\text{N}^+$
16. Which among these is not a structural isomer of the compound C_4H_8
[A] But-1-ene
[B] But-2-ene
[C] But-3-ene
[D] 2-methylpropene
17. Select the incorrect statement regarding alkenes
[A] In alkenes, the carbons are connected by pi bonds
[B] Alkenes have almost same physical properties as that of the alkanes
[C] Alkenes are less reactive than alkanes
[D] Alkenes undergo polymerization reaction
18. Which of the following is a polar aprotic solvent?
[A] DMSO
[B] DMF
[C] Acetonitrile
[D] All of above
19. _____ has more bond length.
[A] N-H
[B] C-H
[C] Cl-Cl
[D] O-H
20. All are the example of Heterogeneous Catalyst. Except One.
[A] Pd
[B] Ni
[C] Lindler
[D] Wilkinson's

SECTION – II

Q 2	Attempt any Two of the following;	10
A	Write reaction, mechanism, reactivity, kinetic and stereochemistry of SN_1 reaction.	05
B	Explain Factor Affecting SN_1 and SN_2 reaction	05
C	Define following terms;	05
	i. Sigma and Pi Bonds	
	ii. Tautomerism	
	iii. Polar and Nonpolar bond	
	iv. Intra and inter molecular forces	
	v. Electrophile and Nucleophile	
Q 3	Attempt any Two of the following;	10
A	Differentiate SP , SP^2 and SP^3 Hybridization.	05
B	Classify different reactive intermediates of carbon and nitrogen with its general formula.	05
C	Construct the structure from the following IUPAC names.	05
	i. 2,2 dimethyl propane	
	ii. 2-Pentanone	
	iii. 2-Methyl propanol	
	iv. Ethanoic acid	
	v. 3-Bromo 3-Methyl Hexane	

SECTION – III

Q 4	Attempt any FOUR of the following;	20
A	Write reaction, mechanism, orientation and reactivity of E_1 mechanism.	05
B	Write any three Preparations and reactions of Alkene.	05
C	What is diene? Classify the various types of diene and explain 1, 4 addition Vs. 1, 2 addition in dienes.	05
D	Write reaction and mechanism for Hoffmann degradation of amide.	05
E	Write any three Preparations and reactions of Alkyl halide.	05
F	Explain Resonance and Inductive effect with suitable example.	05
Q 5	Attempt any FOUR of the following;	20
A	Explain Saytzeff and Markonikov's rule in details.	05
B	Explain physico-chemical properties of amine.	05
C	What is hyperconjugation? Explain it with example.	05
D	Write in detail about the preparation and reaction of Alkynes.	05
E	Write in detail about the preparation and reaction of Alcohol.	05
F	Explain halogenation reaction of Alkane with free radical mechanism.	05